

Finding the prescribers a spend ranking cannot see

An analysis of Colorado Medicare Part D prescribing patterns, 2024

Caleb Edwards, M.D. · Data Analytics Bootcamp Capstone · Front Range Community College / Upright Education · August 2026

Limited hours, thousands of prescribers

Medicare Part D is the prescription drug benefit for people on Medicare. CMS publishes, every year, what every prescriber in the country prescribed and what it cost.

The decision this project supports:

Which prescriber segments should a health system prioritize for outreach, education, or further review — and on what evidence?

The file is public and contains no patient identifiers — no PHI.

Colorado, 2024

\$2.74B

total Part D drug cost

19,390

prescribers

1,177

generic drugs

There are two ways to build a review list



Rank everyone statewide

Sort all 19,390 prescribers by spend, or by cost per prescription, and take the top of the list.

Simple. Fast. It is what most programs do.

But an oncologist and a dentist end up on the same list. The ranking is measuring what kind of medicine you practice, not how you practice it.



Compare each prescriber to their own specialty

Rank each prescriber only against peers in the same specialty, then flag the ones sitting above their own peer group.

Harder to build. Requires the full file.

A family practice clinician is compared to family practice, an oncologist to oncology. Case mix is held roughly constant.

This project tests whether the harder list is worth building.

One file, 390,473 rows, and nothing thrown away

- One row per prescriber, per brand, per generic drug — 22 columns covering claims, 30-day fills, day supply, drug cost, and beneficiary counts.
- 100% of rows are Colorado. The extract is pre-filtered, so no state filter was applied.
- Ten numeric columns loaded as text and were converted to numbers. No value failed to parse.
- Zero duplicate rows, and zero prescribers mapped to two different specialties or two different cities.
- Every blank was left blank. Nothing was imputed, filled, or dropped — blanks in this file are CMS privacy suppressions, not errors.

Reconciliation

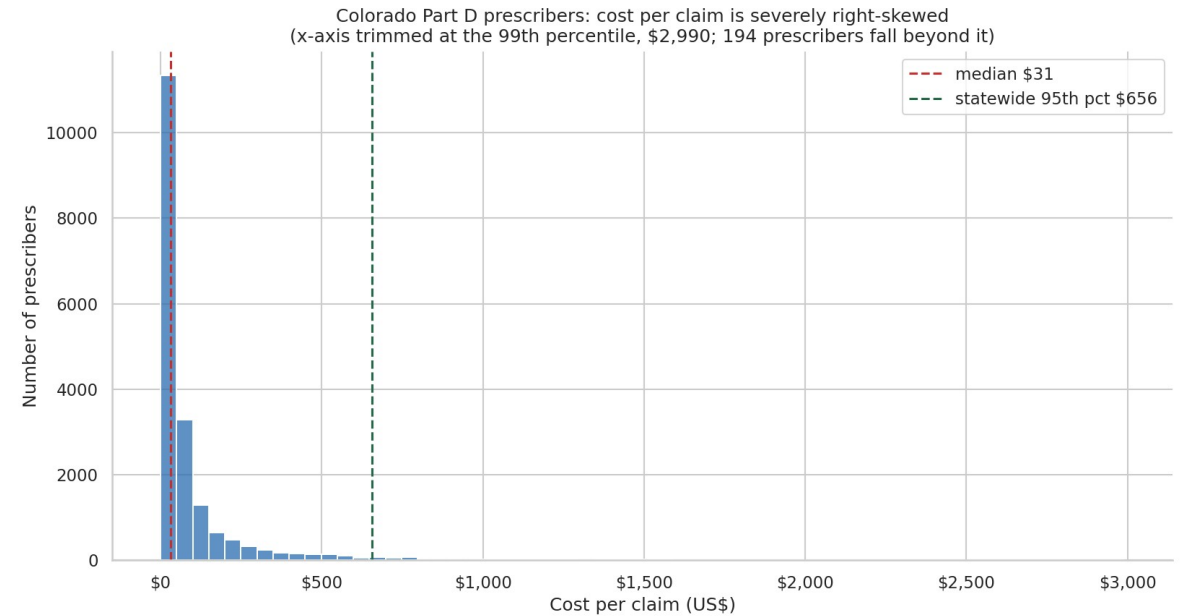
Measure	Raw	Cleaned
Rows	390,473	390,473
Drug cost	\$2.7375B	\$2.7375B
Claims	16,573,710	16,573,710
Prescribers	19,390	19,390
Specialties	97	97

390,473 in, 390,473 out. No delta to explain.

Documented in CLEANING.md — 10 steps, each with a row count before and after.

Cost per prescription, not cost per patient

- Cost per patient would be the natural measure — but the patient-count column is blank on 58.9% of rows under CMS privacy suppression. It cannot be a denominator.
- Claims and drug cost are complete: 0% blank. So the unit metric is cost per claim, with cost per 30-day fill as a companion.
- That distribution is severely right-skewed: median \$31.09, mean \$193.65.
- Skew that severe rules out tests that assume a normal distribution as the primary evidence — which drove the statistics on the next slides.



Four tools, one question, each layer checking the last

1

Excel

Aggregate tables straight from the raw extract. The fast first pass — and the tie-out that every later number is checked against.

2

SQL

The full 390,473-row file as system of record. Percentile window functions (NTILE) build the peer bands and identify outliers inside each specialty.

3

Python

pandas and seaborn. Tests which dimension actually explains cost, ranks prescribers inside their own specialty, and produces the seven charts.

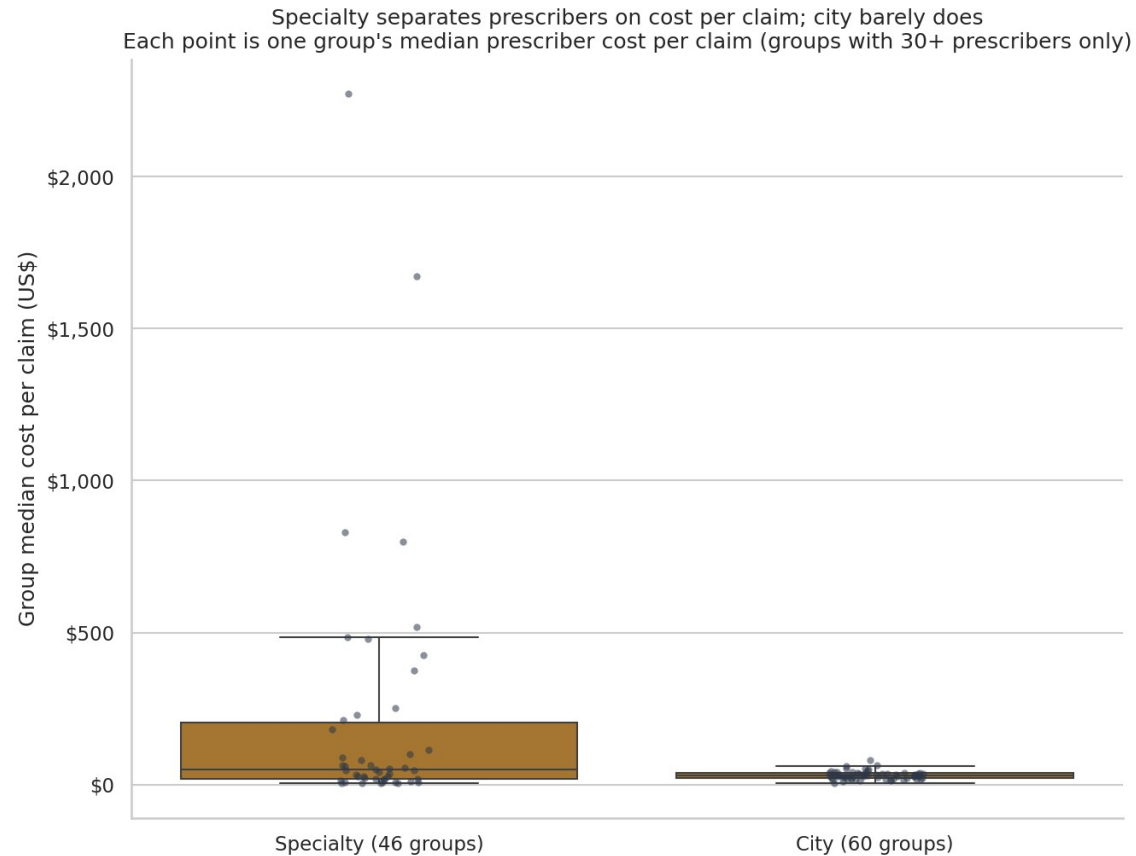
4

Tableau

The executive layer — the prioritized segments in a form a population health director can act on, paired with a one-page memo.

Substitution, stated openly: the proposal called for regression. Regression was never covered in this curriculum, so the same question — which dimension drives cost — is answered with ANOVA, Kruskal-Wallis, and within-specialty percentile ranking.

Specialty drives cost. Geography barely matters.



Specialty

11.2×

spread in median cost per claim between the middle half of the 46 qualifying specialties

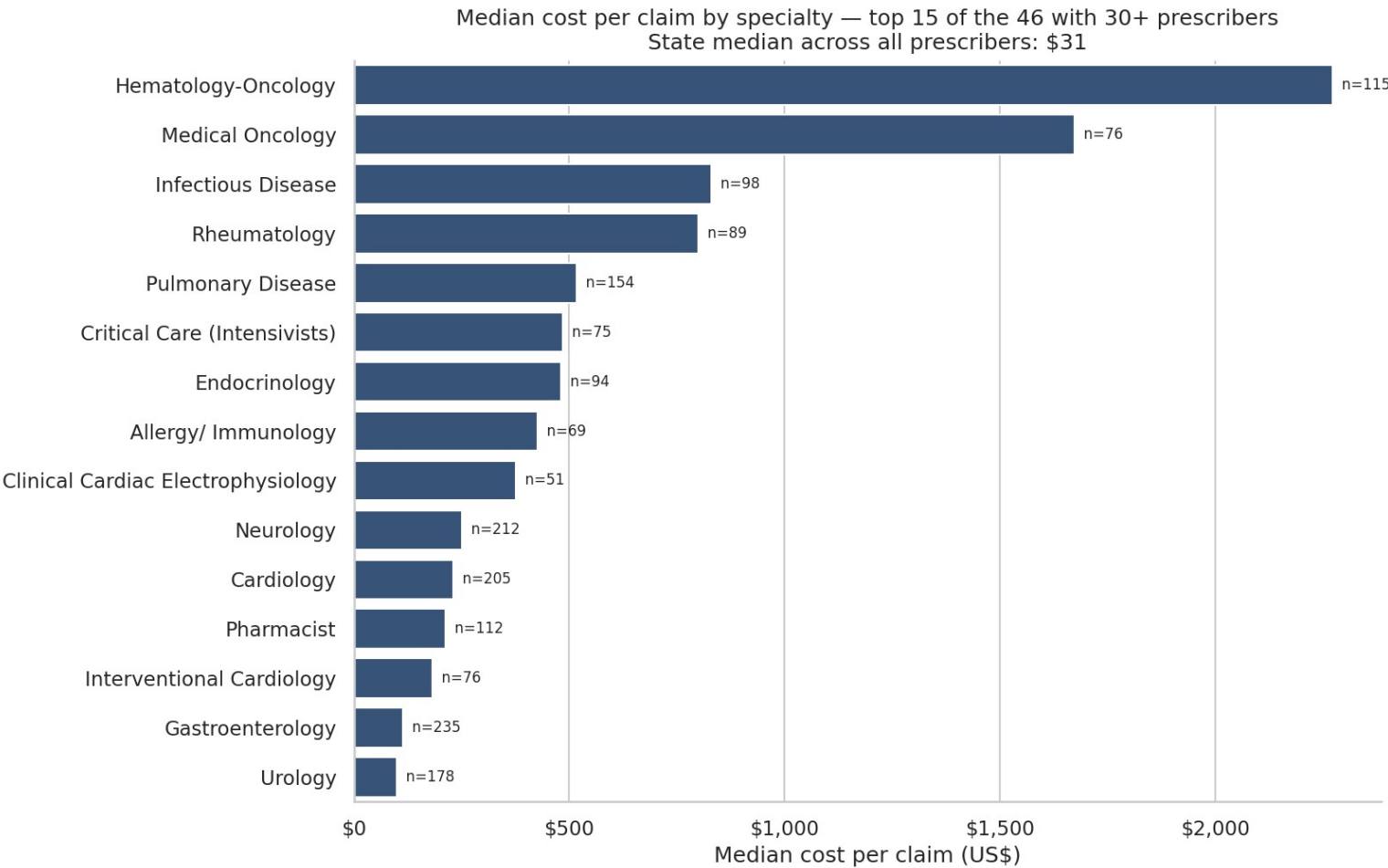
City

1.8×

the same spread across the 60 qualifying Colorado cities

Both dimensions are statistically significant. Only one is practically significant.

One list holds a \$3.89 specialty and a \$2,272 specialty



584×

separates the cheapest qualifying specialty from the most expensive on median cost per claim.

This is case mix, not behavior. Any ranking that ignores it is ranking specialties in disguise.

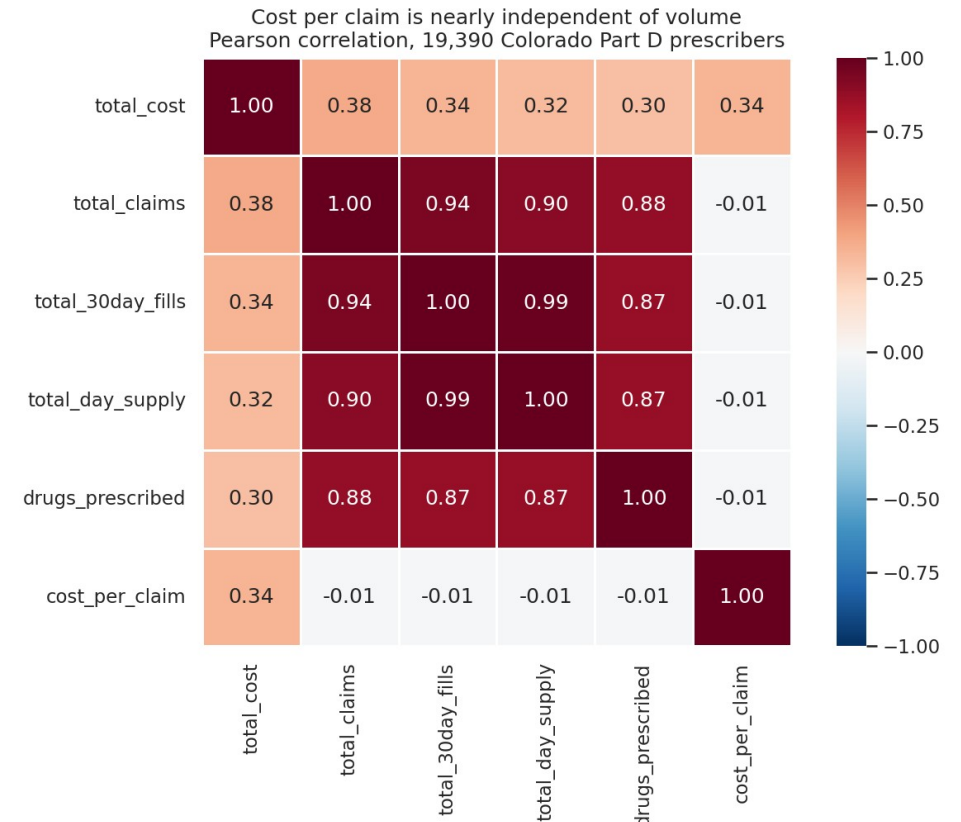
Spending a lot and charging a lot are unrelated

-0.011

correlation between cost per claim and claim volume

A correlation of zero means the two measures carry no information about each other. Knowing how much a prescriber spends tells you nothing about whether their prescriptions are expensive.

So a high-spend review list and a high-unit-cost review list are not two versions of the same list. They are lists of different people.



FINDING 3

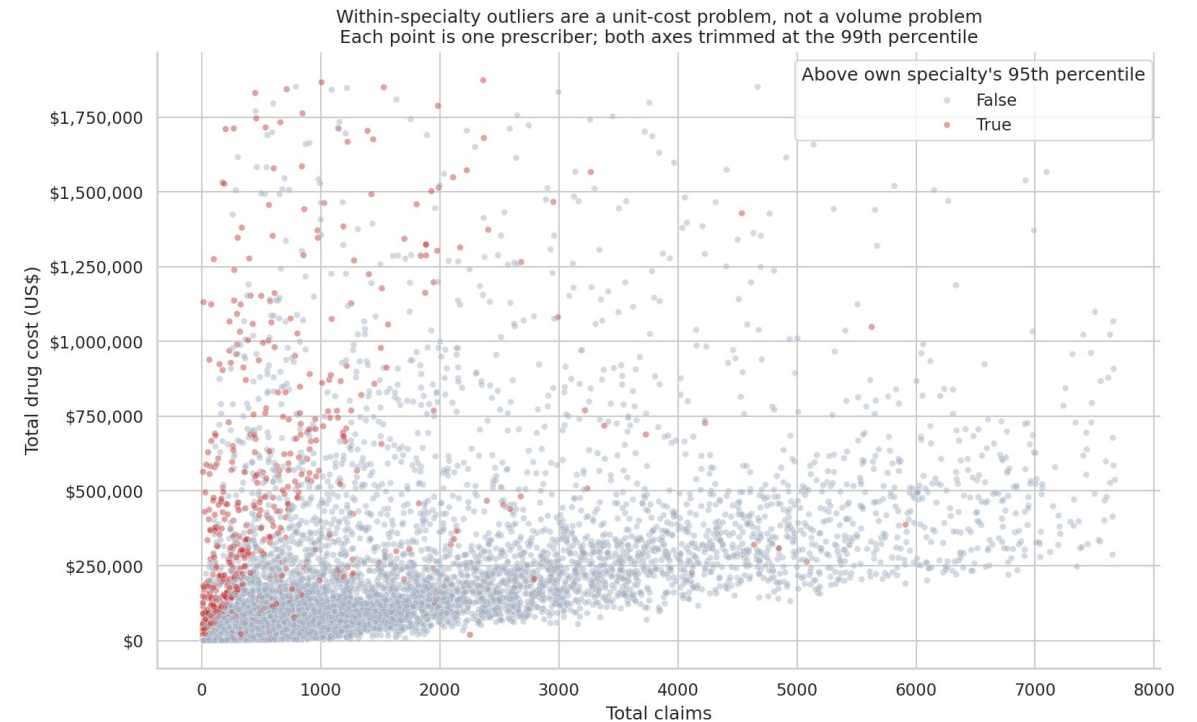
Half the prescribers who stand out among peers are invisible statewide

431 flagged statewide only — high unit cost, but perfectly normal for their specialty.

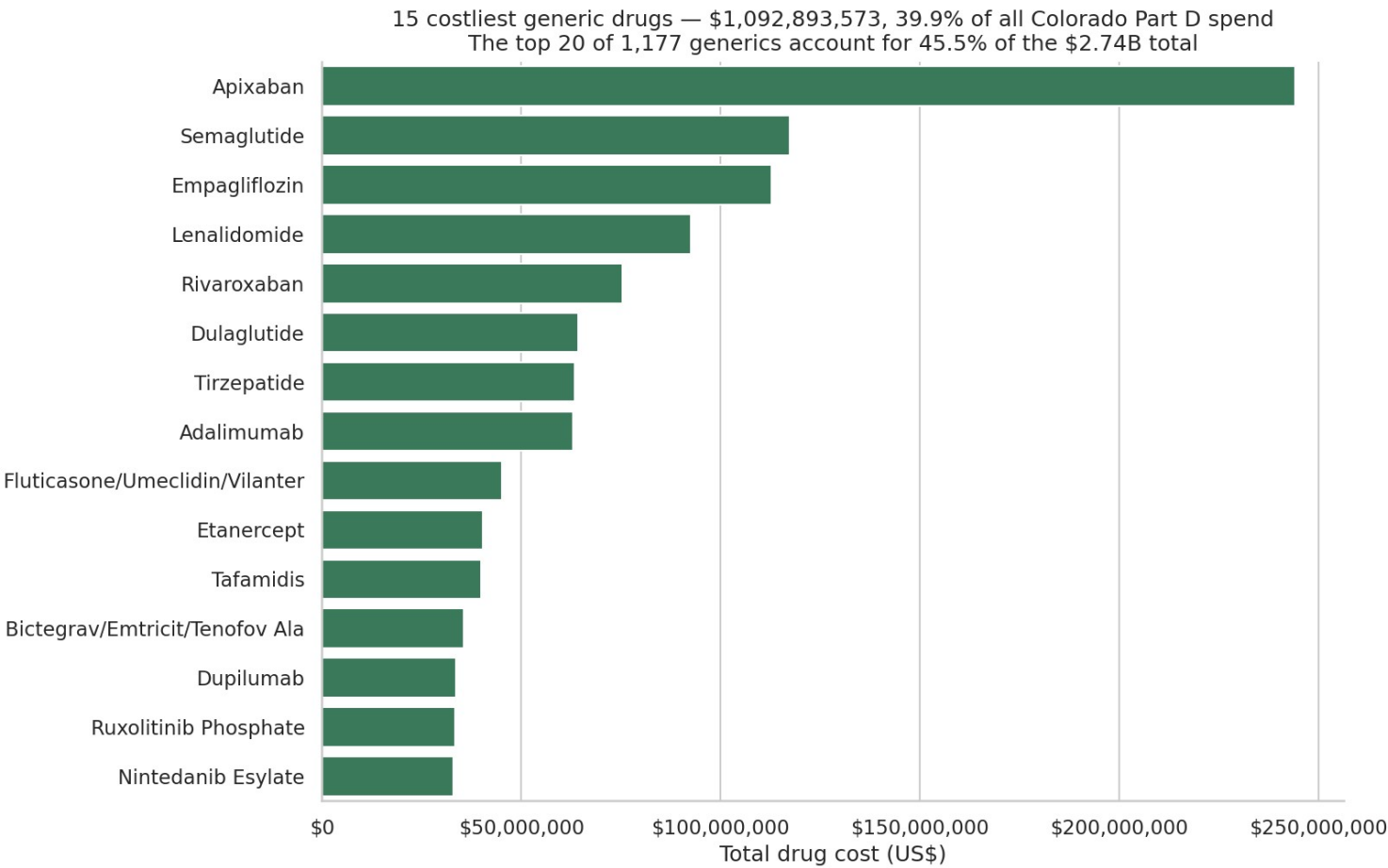
498 flagged by both methods — the overlap.

467 flagged only against their own peers — 48% of the within-specialty list, and they never appear on a statewide ranking at all.

965 prescribers — 5.1% of the state — sit above their own specialty's 95th percentile and carry 22.1% of spend.



The money is concentrated — 20 drugs carry 45% of it



Share of the \$2.74B	
Top 10 drugs	33.5%
Top 20 drugs	45.5%
Top 100 drugs	79.0%
Top 100 prescribers	18.5%
Top 10% of prescribers	71.9%

1.7% of the drug list carries nearly half of Colorado Part D spend. That is what makes a targeted review economically worth doing.

Review these 507 clinicians first — as one cohort

Specialty	Outliers / prescribers	Spend flagged	% of specialty spend	Cost per claim (wtd)
Nurse Practitioner	176 / 3,509	\$109.3M	27.8%	\$137
Physician Assistant	151 / 3,010	\$106.7M	41.6%	\$147
Internal Medicine	65 / 1,292	\$60.3M	19.1%	\$98
Family Practice	115 / 2,288	\$41.3M	11.7%	\$73
Hematology-Oncology	6 / 115	\$33.0M	14.5%	\$2,652
Urology	9 / 178	\$26.7M	53.0%	\$320
Pulmonary Disease	8 / 154	\$25.6M	24.8%	\$907
Cardiology	11 / 205	\$24.1M	17.9%	\$272

Ranked by spend flagged — the 2024 drug cost held by prescribers above their own specialty’s 95th percentile, which is the money a peer-based review would actually be looking at.

\$317.6M

is held by flagged prescribers across the top four segments — 54% of all within-specialty outlier dollars in Colorado, spread across 507 clinicians.

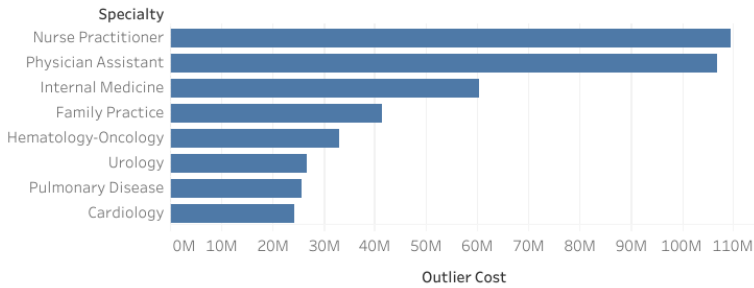
Second tier: Urology

Nine prescribers hold 53% of the entire specialty’s spend — the highest concentration on the list. Nine charts is a day of work.

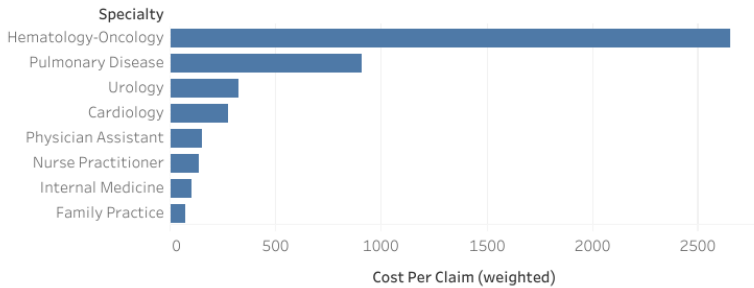
Priority table: FINDINGS.md “Recommended priority segments”. Outlier rule: cost per claim above the prescriber’s own specialty 95th percentile. Cost per claim in this table is specialty-weighted (total cost ÷ total claims), so it runs higher than the prescriber-level medians on Finding 1.

The same finding, in the view a director would open

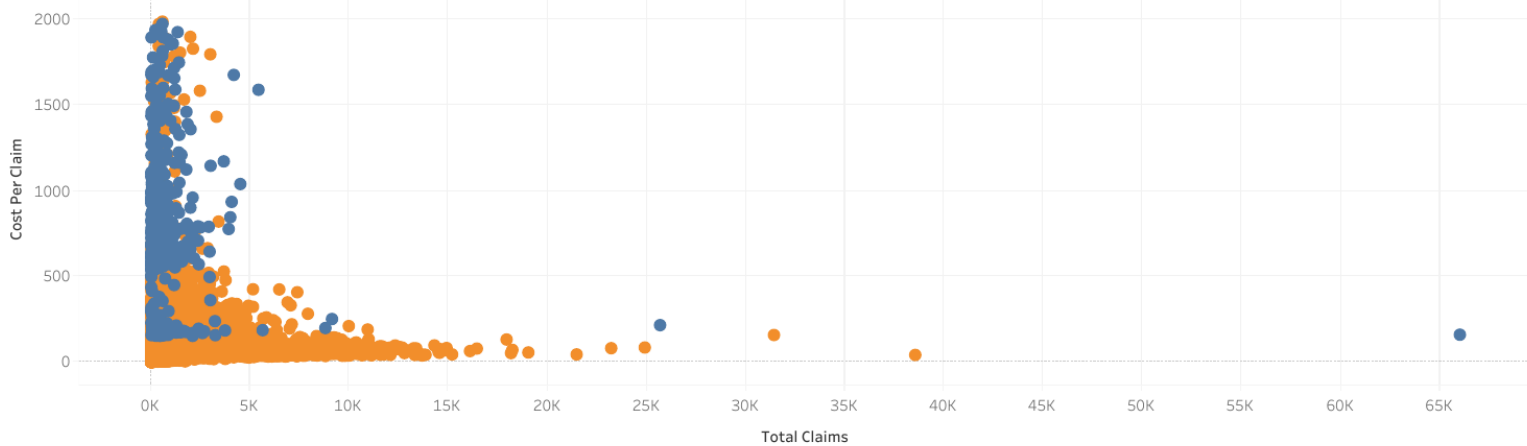
Priority Segments



Unit Cost By Segment



Prescriber Scatter



1st vs 6th

Nurse Practitioner is first on flagged spend and sixth on cost per claim. Hematology-Oncology is fifth and first.

Almost inverted. In the scatter, blue marks the 965 prescribers above their own specialty's 95th percentile.

What this analysis does not say

1 No clinical context

The file has no diagnoses, no severity, no panel composition. An outlier here is a statistical outlier — a reason to look, never a finding of waste or wrongdoing.

2 Deprioritize the expensive specialties

Oncology, pulmonary and cardiology rank high on dollars because their drugs genuinely cost more. The residual peer-level signal is real but small, and the clinical justification is stronger.

3 Cost is gross, not net

Tot_Drug_Cst is the gross point-of-sale price — what the plan, the patient and third parties paid combined, before manufacturer rebates — so cross-drug comparisons overstate the spread on rebated brands.

4 One year, one state, and 51 specialties excluded

2024 Colorado only — no trend, no national benchmark. The ≥ 30 -prescriber rule drops 51 of 97 specialties, covering 2.3% of prescribers and 3.0% of spend.

What I would do next

1

Run the 507-clinician cohort as a pilot

Peer-comparison letters and pharmacist outreach, then measure whether the peer gap narrows in the next data year.

2

Scope a formulary review to 20 drugs

Those twenty carry 45.5% of state spend — the cheapest structural lever available.

3

Rebuild the flag with clinical context

Diagnoses and panel composition would turn a statistical outlier into an actionable one. That data lives inside a health system, not in a public file.

4

Repeat on the 2023 file

One year cannot separate a persistent pattern from a single unusual year. Two can.

Thank you — questions welcome.

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